

Academic Performance and Inequality: Can Generative AI Widen the Digital Divide in Higher Education?



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1 Introduction

Over the past decade, higher education has undergone a rapid transformation driven by digitalization, the growth of online learning, and the development of artificial intelligence (AI) technologies [1]. This evolution has accelerated since 2022 with the emergence of advanced generative models such as ChatGPT, which have introduced new dynamics in the production and access to knowledge [2]. These technologies, based on large language models (LLMs), are capable of generating textual, visual, auditory, and computational content in a coherent, creative, and contextually appropriate manner [3, 4].

Generative artificial intelligence (GAI) differs from other branches of AI due to its ability to create original content, such as essays, images, code, and explanations, based on human instructions. Unlike models focused on prediction or classification, generative models produce novel responses grounded in previously learned linguistic and contextual patterns [5]. Their use in higher education has rapidly expanded through tools such as ChatGPT, DALL·E, Midjourney, and GitHub Copilot, which support writing processes, concept visualization, problem-solving, and code development [6–8].

This landscape has opened unprecedented opportunities to personalize learning, foster student autonomy, and optimize teaching processes. However, its implementation has also introduced a new field of pedagogical, institutional, and ethical tensions.

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Various studies warn that effective adoption of these technologies is not equally accessible to all: access to devices, Internet connectivity, digital literacy, and institutional conditions are key factors in how GAI impacts academic performance [9–12]. For instance, recent data (UNESCO, 2023) indicates that in Latin America, only 27% of students at rural universities have access to AI-compatible devices, and fewer than 40% have stable Internet connection for continuous use.

In this context, GAI, far from being a neutral technology, may become a new source of academic differentiation if its use turns into an implicit competency mastered only by some students. Educational inequality thus shifts from access to technology to the meaningful and critical use of it, generating emerging forms of technical, cognitive, and cultural exclusion [13].

From a critical perspective, this article argues that the integration of GAI in higher education must be analyzed beyond its technical or innovative dimensions. It is essential to understand it as part of a socio-educational framework marked by structural asymmetries, where technology not only reflects but can also amplify existing inequalities. The discussion about its impact must consider not only access conditions, but also pedagogical practices, institutional frameworks, and cultural repertoires that shape its use [14, 15].

Based on this perspective, the analysis proposed here adopts a critical narrative review that seeks to identify how GAI interacts with dynamics of equity, quality, and educational justice in Latin American higher education. This approach allows for a better understanding of the factors that influence its implementation, as well as the risks and opportunities that arise in contexts of social and technological exclusion [16, 17].

Accordingly, this article is based on the hypothesis that, if not implemented equitably and contextually, GAI can deepen existing inequalities in higher education, particularly affecting students in vulnerable situations. From this standpoint, the potential of generative artificial intelligence to influence academic performance is examined, considering both its benefits and the risk that it may become a new factor widening the digital divide at this level of education.

2 Materials and Methods

This study adopts a critical narrative review approach, with an exploratory and positive character, aimed at examining the implications of GAI in higher education, particularly in relation to academic performance and the digital divide. The target context of analysis is university-level teaching and learning, with a focus on both students and faculty as the main actors affected by the integration of generative AI in academic settings.

Unlike systematic reviews or meta-analyses, which require methodological homogeneity and quantifiable results, a critical narrative review is more suitable in contexts where the field of study is recent, diverse, and undergoing constant transformation. In this case, the emerging nature of GAI and the plurality of pedagogical, ethical, and

technical perspectives associated with its educational use justify the application of an interpretative, flexible, and context-sensitive approach. This allows for the integration of not only empirical evidence but also relevant theoretical frameworks and qualitative analyses [15, 18].

2.1 Selection Criteria

- The following criteria were established for the construction of the documentary corpus:
- Time frame: Publications from 2020 to 2024 were selected, corresponding with the rise of generative tools such as ChatGPT, DALL-E, and Copilot in academic settings.
- Type of documents: Included materials comprised scientific articles, academic reviews, case studies, indexed theoretical essays, and technical reports relevant to the fields of higher education and artificial intelligence.
- Information sources: The search was conducted through specialized databases such as Scopus, Web of Science, SpringerLink, MDPI, IEEE Xplore, and Frontiers in Education. Google Scholar was also used to access gray literature and key open-access documents.

2.2 Analyzed Corpus

From an initial pool of over 40 identified documents, 18 were selected for the final analysis: 6 empirical studies, 5 theoretical reviews, 3 academic essays, and 4 technical reports. This diversity enabled the contrast of pedagogical, technological, and socio-ethical approaches, enriching the critical analysis.

2.3 Analysis Procedure

The methodology was developed in three successive phases. First, an exploratory reading and pre-selection were carried out, in which more than forty relevant documents were identified. This initial stage considered thematic relevance, methodological rigor, and explicit focus on higher education contexts.

Second, the process of thematic coding was conducted through open reading and inductive categorization. This allowed the identification of patterns and emerging categories grouped around five central axes: access and critical appropriation of generative AI, impact on academic performance, digital and cultural inequalities, teacher roles and AI training, and practical applications with their ethical limitations.

Finally, a categorical synthesis and critical analysis were performed, linking the thematic categories with the conceptual framework of structural inequality. This integrative perspective made it possible to recognize tensions, ambivalences, and opportunities in the educational use of generative AI, while highlighting the ethical and pedagogical implications of its unequal adoption.

Overall, this approach facilitated the mapping of current knowledge, the identification of gaps in literature, and the proposal of future lines of research oriented toward technological equity, teacher development, and educational justice within university contexts affected by digital inequality.

3 Emerging Challenges: Inequality and the Impact of GAI in Higher Education

One of the main challenges in integrating GAI into higher education is the unequal access to technological infrastructure, which directly affects students' actual opportunities for participation and academic benefit. Although tools such as ChatGPT, Copilot, and DALL-E are promoted as being accessible from any Internet-connected device, their effective use requires certain minimum conditions, such as stable connectivity, up-to-date hardware, and technologically equipped institutional environments, which are not guaranteed across all university settings [19–21].

This gap goes beyond physical access to devices or networks; it extends to the conditions required for sustained use, particularly in rural areas, under-resourced institutions, or countries in the Global South. In these contexts, many students must share family-owned devices, cope with unstable Internet connections, or rely on limited versions of these tools [22, 23]. Moreover, universities with lower levels of investment in digital transformation often lack clear policies, technical support, and faculty training, further widening the asymmetry between institutions. While some are exploring automated tutoring systems or AI-generated curricular content, others struggle to provide even basic connectivity in their classrooms.

In this context, GAI runs the risk of reinforcing existing exclusionary dynamics, wherein students with greater technological resources not only gain easier access to these tools, but also develop a more fluid, productive, and strategic relationship with them. This results in a growing academic performance gap—not due to differences in students' capabilities, but rather as a consequence of structural inequalities in access to and use of technology [24].

3.1 Differential Digital Literacy

Beyond material access to technologies, the effective adoption of generative AI tools in higher education requires a complex set of cognitive, technical, and critical

competencies. This differentiated digital literacy is reflected not only in familiarity with platforms, but also in mastery of academic language, the ability to formulate effective prompts, and the skill to critically evaluate AI-generated content [25, 26]. Recent studies highlight that productive interaction with generative models, such as ChatGPT or Copilot, depends on knowing how to pose questions, how to interpret responses, and how to reuse information critically. This skill, known as prompt engineering, demands structured thinking, conceptual mastery, and an understanding of the tool's limitations. Students with higher cultural capital or prior training in digital competencies tend to hold a significant advantage.

Moreover, there is a gap in the ability to distinguish between original, verified, and replicated content. Many students accept AI-generated responses as valid without questioning their accuracy or bias, which can reinforce misinformation or reproduce stereotypes [24]. Inequality in digital literacy not only affects the pedagogical effectiveness of GAI, but also directly impacts learning quality, favoring those who not only have access to the tool but also know how to use it strategically and assess its outputs. This cognitive gap constitutes a new form of academic exclusion [23].

3.2 Uncritical Dependence Versus Meaningful Appropriation

A central pedagogical dilemma in the integration of GAI into higher education lies in the tension between uncritical use and meaningful appropriation. Although these technologies offer significant potential to enrich learning processes—by facilitating explanations, drafts, or feedback, their decontextualized use can foster passive, imitative behaviors disconnected from critical thinking. This may lead to a growing “algorithmic dependence” in academic settings that prioritize speed of output over the depth of cognitive engagement [23, 24].

The ability to obtain immediate answers without researching, reworking, or cross-referencing information tends to reduce intellectual effort and erode essential competencies such as argumentation, critical analysis, and academic synthesis. In contrast, some innovative educational environments advocate for a meaningful appropriation of GAI, where these tools serve as mediators of learning rather than substitutes for thought. From this perspective, the use of GAI must be accompanied by teacher mediation that encourages ethical and reflective integration, guiding students to develop metacognitive skills that enable them to question, evaluate, and rework the generated content [17].

3.3 Inequitable Assessment

The expansion of GAI in higher education has raised concerns regarding the equity and validity of assessment processes. As students increasingly incorporate tools like ChatGPT or Claude into their work, it becomes challenging to discern which portions

are original and which are AI-assisted. This has intensified dilemmas surrounding the authenticity of learning and academic authorship. Some students openly declare their use and integrate the technology critically; others use it as a substitute for intellectual effort without disclosure. This ambiguity results in unequal treatment, favoring those who manipulate the tools more skillfully and disadvantaging those who are transparent but less digitally sophisticated [24].

Moreover, traditional assessment instruments are not designed to distinguish between acquired knowledge and AI-assisted knowledge, which may perpetuate biases related to social class and digital competence. A redefinition of assessment criteria is necessary, focusing on process, reflection, and the ethics of learning.

3.4 Reproduction of Structural Inequalities

Generative artificial intelligence can become a reproducer of structural inequalities if it is not implemented with an inclusive approach. Although it promises to democratize access to knowledge, its effective use depends on conditions that are not equal for all. Students from socioeconomically vulnerable backgrounds face both material and symbolic barriers to accessing and appropriating these technologies. This includes lack of devices, unstable connectivity, limited technological familiarity, and scarce training in AI-mediated academic discourse [21].

In better-resourced institutions, competencies in prompting and creative use of GAI are developed. Conversely, students in disadvantaged environments lack sufficient training and faculty support, which crystallizes a scenario of accumulated advantages. The absence of institutional policies for leveling and inclusive training reinforces an exclusionary meritocratic logic. Thus, GAI could become a new gatekeeper of exclusion if it is not embedded in ecosystems committed to equity and the right to learn with technology.

4 Current Applications of GAI and Its Impact on Academic Performance

In the context of higher education, GAI has given rise to a series of concrete applications that are transforming the way students learn, produce knowledge, and engage with academic content. When integrated across various university activities, these tools have demonstrated potential both to enhance academic performance and to promote new forms of teaching and assessment. Below is a synthesis of the main current applications of GAI in educational settings, highlighting their pedagogical utility and the academic references supporting their impact (Table 1).

Table 1 Current applications of GAI

GAI application	Description	Source
Generation of summaries, essays, and personalized presentations	Large language models (LLMs) such as ChatGPT, Claude, and Gemini enable the generation of structured academic texts from brief prompts. This facilitates comprehension, synthesis, and organization of information, promoting autonomy and critical thinking when accompanied by teacher supervision	[21, 22]
Conversational assistants as virtual tutors	Models like ChatGPT and Bing Chat act as virtual tutors by providing personalized explanations adapted to the student’s pace. They improve self-efficacy and autonomy, although their effectiveness depends on the level of digital literacy and pedagogical framework	[17, 18, 21, 22]
Automatic feedback on exercises or writings	Tools such as GPT-4 and Claude analyze student texts and offer immediate suggestions regarding writing, cohesion, and argumentation. They also support practical exercises and foster metacognitive skills, though pedagogical mediation is needed to avoid passive dependence	[14, 22, 24, 25]
Code generation and mathematical problem-solving in STEM fields	Tools like Copilot and Codex assist programming and technical problem-solving. They can write code, explain processes, and facilitate simulations, although without guidance they may cause operational dependence	[14, 25]
Exam simulation and automatic analysis of legal or scientific texts	These tools enable the preparation of adaptive assessments and the analysis of complex texts (legal, scientific). They foster critical comprehension, reduce evaluative anxiety, and personalize learning, but require pedagogical validation to avoid errors or biased interpretations	[21, 22, 25]

5 Strategies for Inclusive Use of GAI

5.1 Promotion of Equitable Institutional Environments

The incorporation of GAI in higher education presents significant opportunities, but also critical challenges which, if not managed with an inclusive approach, may deepen existing gaps (Table 2). Below is a structured proposal based on five strategic axes that integrate both solutions and the challenges identified in previous chapters.

Table 2 Challenges and solutions

Identified challenges	Proposed strategies	Source
Unequal access to technological infrastructure	Promotion of equitable institutional environments: provision of devices, connectivity, and digital spaces	[13, 18–20]
Gap in critical digital literacy	Faculty and student training: modules on prompting, ethics, biases, and AI analysis in each discipline	[23, 25]
Uncritical dependence on AI	Pedagogical models: hybrid tutoring, teacher mediation, and metacognitive use of GAI	[25]
Inequitable assessments and lack of adaptive criteria	Context-sensitive adaptive assessments with teacher validation and formative feedback	[21, 22]
Reproduction of structural inequalities	Inclusive curricular design with transversal focus, contextualized practices, and respect for diverse trajectories	[17, 18, 23, 24]
Absence of inclusive public policies	Development of research lines on equitable policies and regulatory frameworks	[10]

5.2 Institutional Policies and Infrastructure

The adoption of GAI in the university context requires clear institutional policies that guarantee equitable access conditions. Institutions must provide basic technological infrastructure (devices, connectivity, and digital spaces) and establish technical and pedagogical support mechanisms for students with non-traditional educational backgrounds or socioeconomically vulnerable situations.

Furthermore, guided training spaces, technological tutoring, and support strategies for the critical use of AI should be implemented. As noted by [21], the success of these tools depends not only on their technical interface but on the educational ecosystem that supports them. This infrastructure must be accompanied by an ethical, participatory, and regulatory institutional culture, including protocols for use, academic integrity, and data protection incorporated into institutional digital policies.

5.3 Teacher and Student Training in Critical AI Literacy

Responsible use of GAI requires critical literacy for both faculty and students. This training must go beyond technical proficiency to include understanding biases, ethical limitations, and the transformative potential of these technologies. For educators, three key competencies are identified: (1) pedagogical design with GAI; (2) contextualized evaluation of generated content; and (3) reflective guidance of AI-supported learning.

For students, literacy should focus on developing skills in prompting, data verification, bias identification, and differentiating between AI-assisted creativity and plagiarism. The inclusion of specific modules in undergraduate, graduate, and continuing education curricula enables systematic and transversal integration. These modules should cover the functioning of large language models (LLMs), bias analysis, ethical practices, and applied cases according to each discipline [17].

5.4 Inclusive Teaching Models with AI

A pedagogical design is proposed in which GAI functions as a cognitive mediator rather than a teacher replacement. This approach promotes the development of critical thinking and student autonomy. The hybrid tutoring model, combining human and AI feedback, allows for a personalized learning experience that respects students' cognitive and sociocultural diversity. For example, tools like ChatGPT can facilitate access to resources, while the teacher fosters metacognitive processes and validates the generated information.

Additionally, the implementation of context-sensitive adaptive assessments is proposed, which can adjust to the student's pace, level, and conditions. These assessments should incorporate algorithmic transparency and teacher validation to avoid biases or exclusionary automatisms [26].

5.5 Inclusive and Flexible Curriculum

The integration of GAI demands a cross-cutting curricular transformation. This is not about reserving these tools for technological programs but rather democratizing their access across all academic disciplines. For instance, GAI can be adapted to generate study guides in pedagogy, perform text analysis in law, or create clinical simulations in medicine.

The curriculum should strengthen competencies such as digital literacy, computational thinking, and metacognition, incorporating content on ethics, technical functionality, and contextualized application. This can be achieved through elective modules, cross-disciplinary content, or extracurricular workshops, respecting the diversity of student trajectories and avoiding the imposition of uniform technological standards [24].

5.6 *Future Research Lines*

The development of GAI in education opens new lines of critical research:

- Correlation between GAI use and academic performance in marginalized contexts: It is necessary to study whether these tools act as compensatory mechanisms or amplify inequalities. Mixed methodologies are suggested to measure variables such as performance, motivation, retention, and self-efficacy in vulnerable groups.
- Inclusive public policies for GAI integration: Research should investigate comparative models of technological inclusion in countries with high inequality, analyzing successful practices, risks, and contextual adaptation mechanisms. Additionally, monitoring indicators and co-creation mechanisms with educational stakeholders should be developed to ensure relevant and equitable technology policies.

International institutional examples: Universities such as Stanford (USA), Aalto (Finland), and the Autonomous University of Barcelona have begun integrating AI literacy modules and developing inclusive policies for implementation, serving as references for designing strategies adapted to the Latin American context [24]. These strategies not only enable informed use of GAI but also ensure that its implementation contributes to closing gaps, strengthening educational justice, and promoting more meaningful and democratic learning.

6 Conclusions

Generative artificial intelligence holds great potential to transform higher education, but it also presents risks if not implemented with an inclusive approach. Its use can enhance learning and academic production, yet it may also widen existing inequalities. Furthermore, unequal access to technology, lack of critical training, and absence of clear policies hinders all students from benefiting equally. This generates new forms of exclusion, especially for those from less privileged backgrounds.

Despite its advantages, such as personalized learning and immediate feedback, GAI is not inherently beneficial. Its positive impact depends on teacher mediation, institutional support, and critical digital literacy. In this context, to prevent GAI from exacerbating the educational gap, five key strategies are proposed: ensuring equitable access, training faculty and students, adapting pedagogical models, transforming curricula, and promoting inclusive public policies.

Acknowledgements The authors employed AI-assisted tools (ChatGPT 5) for stylistic and grammatical refinement. Full responsibility for the final content remains with the authors.

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